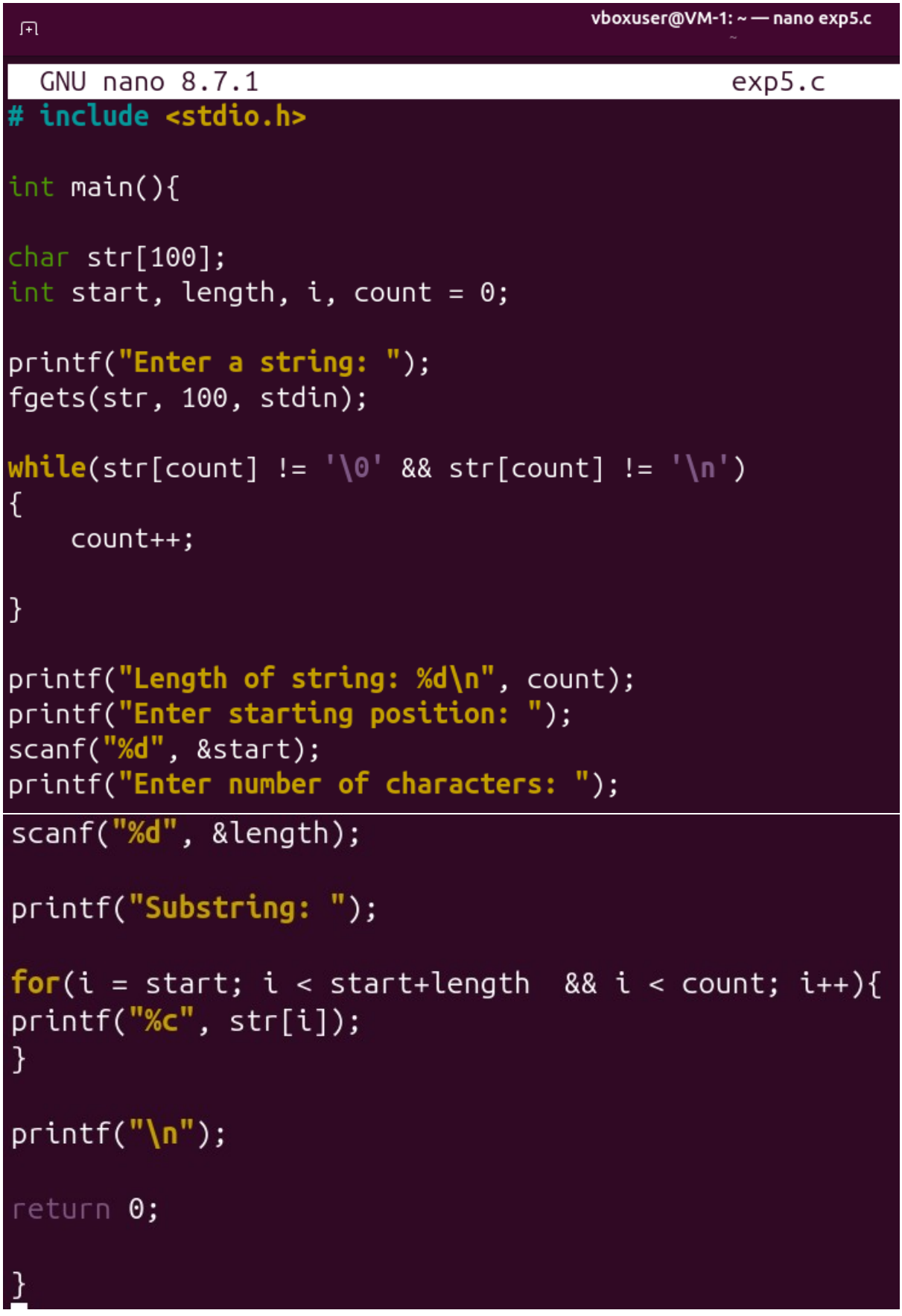
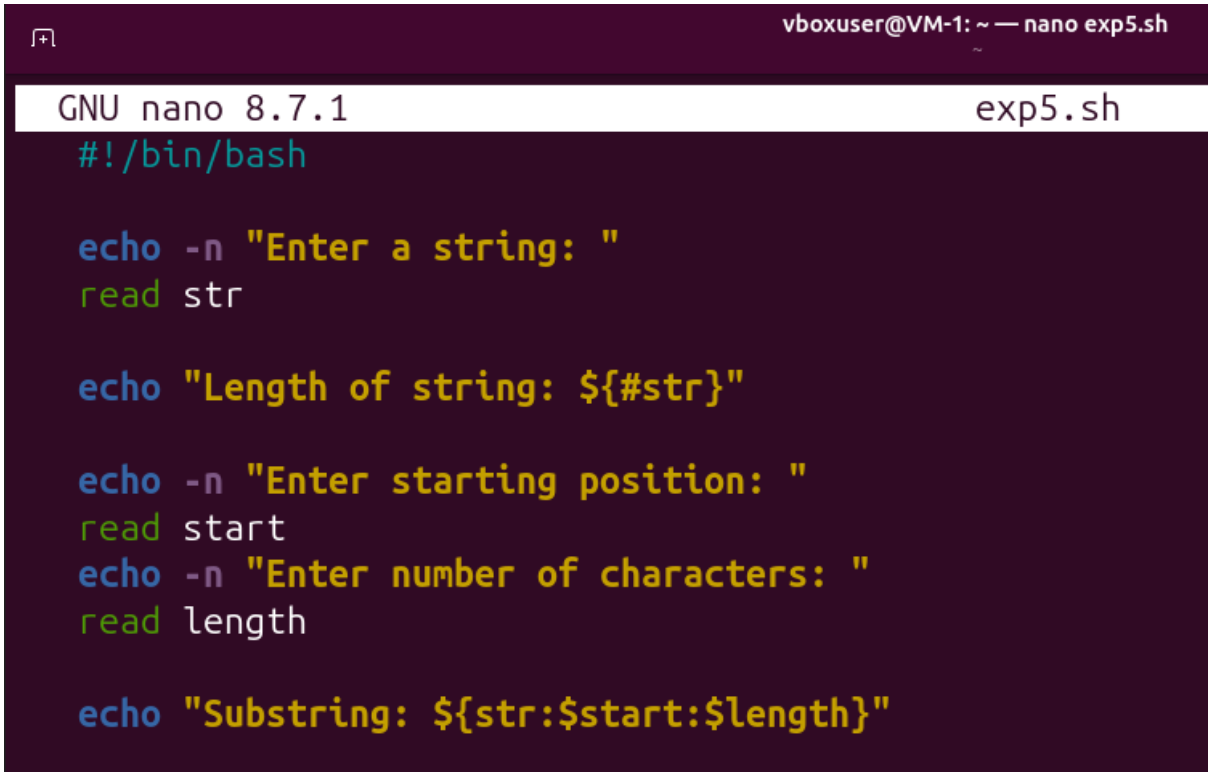
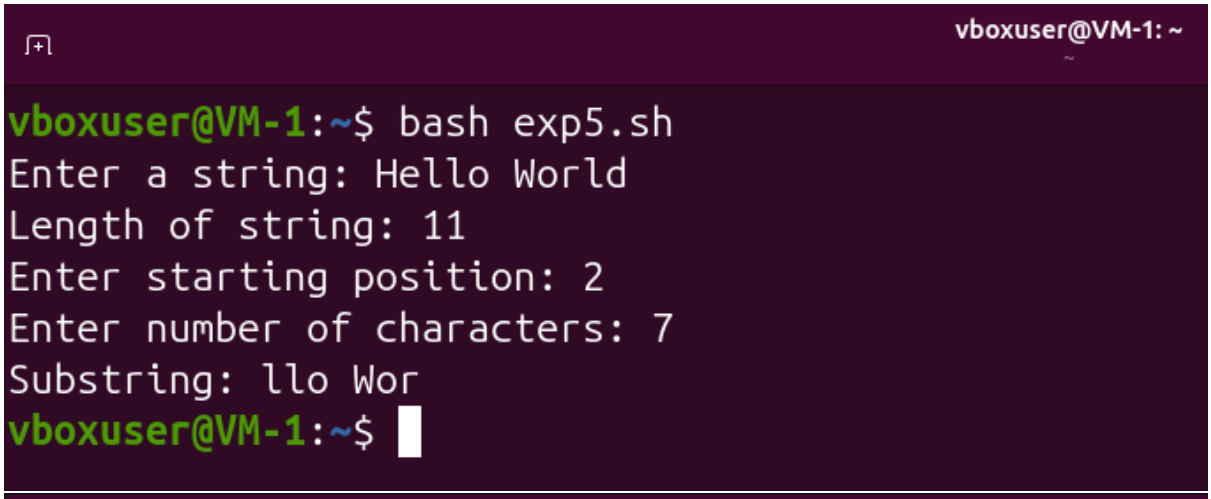
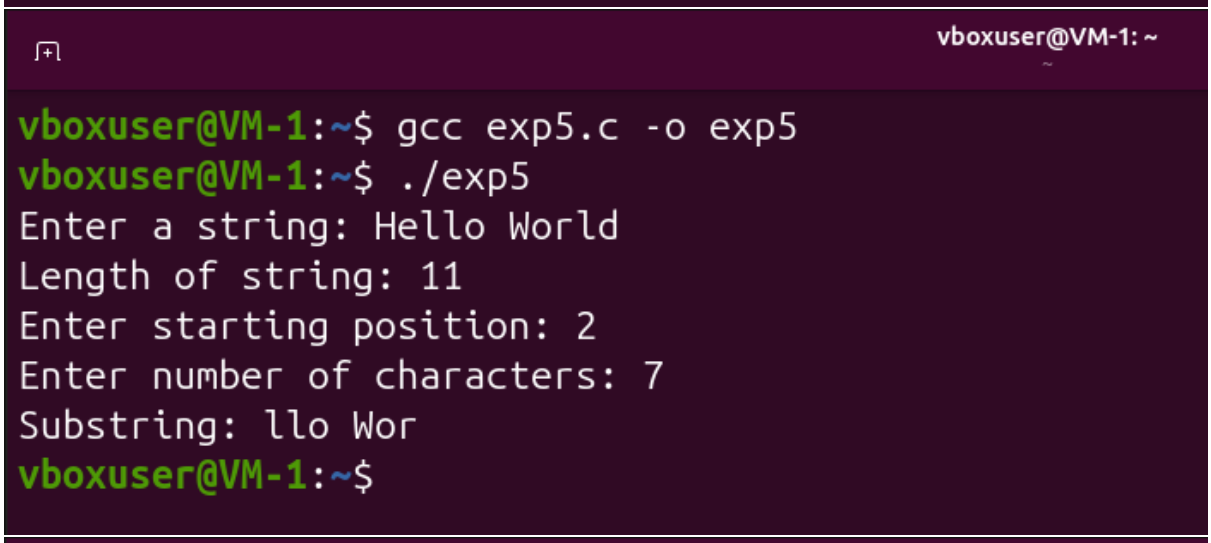
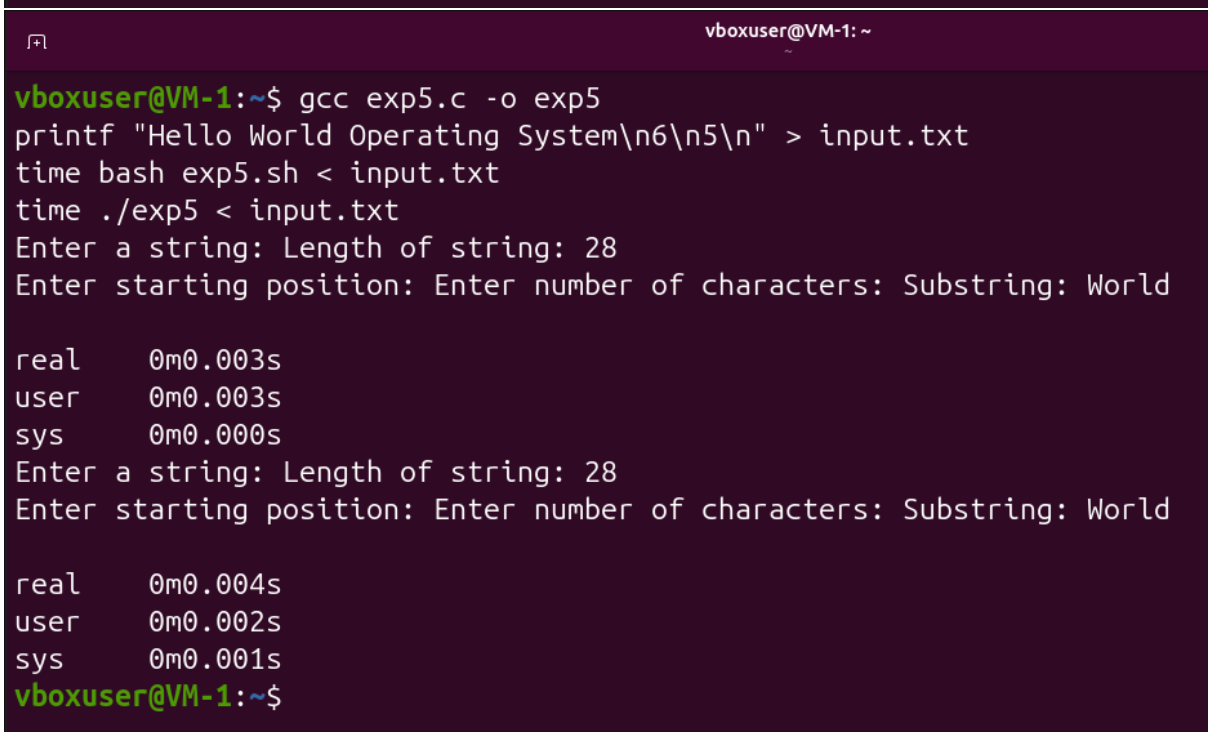

vboxuser@VM-1: ~ — nano exp5.c
GNU nano 8.7.1 exp5.c
include <stdio.h>
int main(){
char str[100];
int start, length, i, count = 0;
printf("Enter a string: ");
fgets(str, 100, stdin);
while(str[count] != '\0' && str[count] != '\n')
{
count++;
}
printf("Length of string: %d\n", count);
printf("Enter starting position: ");
scanf("%d", &start);
printf("Enter number of characters: ");
scanf("%d", &length);
printf("Substring: ");
for(i = start; i < start+length && i < count; i++){
printf("%c", str[i]);
}
printf("\n");
return 0;
}


vboxuser@VM-1: ~ — nano exp5.sh
GNU nano 8.7.1 exp5.sh
#!/bin/bash
echo -n "Enter a string: "
read str
echo "Length of string: ${#str}"
echo -n "Enter starting position: "
read start
echo -n "Enter number of characters: "
read length
echo "Substring: ${str:$start:$length}"


vboxuser@VM-1: ~
vboxuser@VM-1:~$ bash exp5.sh
Enter a string: Hello World
Length of string: 11
Enter starting position: 2
Enter number of characters: 7
Substring: llo Wor
vboxuser@VM-1:~$


vboxuser@VM-1: ~
vboxuser@VM-1:~$ gcc exp5.c -o exp5
vboxuser@VM-1:~$ ./exp5
Enter a string: Hello World
Length of string: 11
Enter starting position: 2
Enter number of characters: 7
Substring: llo Wor
vboxuser@VM-1:~$


vboxuser@VM-1: ~
vboxuser@VM-1:~$ gcc exp5.c -o exp5
printf "Hello World Operating System\n6\n5\n" > input.txt
time bash exp5.sh < input.txt
time ./exp5 < input.txt
Enter a string: Length of string: 28
Enter starting position: Enter number of characters: Substring: World
real 0m0.003s
user 0m0.003s
sys 0m0.000s
Enter a string: Length of string: 28
Enter starting position: Enter number of characters: Substring: World
real 0m0.004s
user 0m0.002s
sys 0m0.001s
vboxuser@VM-1:~$